

Basic Electrical Engineering

MODULE 2 — PART 2: QUESTION BANK

Module 2 (10 hrs): AC Circuits

For First-Year Students — with complete answers and diagrams

Course: Basic Electrical Engineering · Course Code: EE100

Question Paper Structure

Section	Marks per Question	Scope	Questions
Section A	2 marks	Compulsory — from every module	Q1 - Q34
Section B	5 marks	From Modules 1 and 2 only	Q1 - Q16
Section C	7 marks	Compulsory — from every module	Q1 - Q10
Section D	Numericals	Solved problems (all topics)	N1 - N12

- Every section covers **all the topics** of Module 2 — no topic is left out.
- No question is repeated between sections.
- Section D gives every numerical problem solved step by step, with its circuit or phasor diagram.

SECTION A — 2-Mark Questions

Q1. What is an alternating voltage?

Answer: A voltage which periodically changes its polarity at regular intervals of time is called an alternating voltage. Its magnitude and direction both vary with time.

Q2. Why is a sine waveform used for AC?

Answer: It produces the least disturbance, is the smoothest and most efficient waveform, produces a revolving magnetic field that can do work, and makes the mathematics simple (any periodic wave can be built from sinusoids).

Q3. Define cycle, time period and frequency.

Answer: A cycle is one complete set of positive and negative values. The time period T is the time for one cycle. Frequency f is the number of cycles per second (Hz), $f = 1/T$.

Q4. Define the instantaneous value and amplitude of an alternating quantity.

Answer: The instantaneous value is the value at any instant (v or i). The amplitude (peak value) V_m is the maximum value, positive or negative, attained by the quantity.

Q5. Write the relation between angular velocity and frequency.

Answer: $\omega = 2\pi/T = 2\pi f$, where ω is in radians per second.

Q6. Write the general equation of a sinusoidal voltage.

Answer: $v(t) = V_m \sin(\omega t + \phi)$, where V_m is the amplitude, ω the angular velocity and ϕ the phase.

Q7. What is meant by lead and lag?

Answer: If one sinusoid reaches its maximum before another, it is said to lead; the other lags. If the phase difference is zero they are in phase, otherwise they are out of phase.

Q8. What is the peak-to-peak value?

Answer: The peak-to-peak value is the difference between the positive and negative peaks, equal to $2V_m$.

Q9. What is the average value of an alternating quantity?

Answer: The average over a full cycle is zero. The half-cycle average is $0.637 V_m$ (i.e. $2V_m/\pi$).

Q10. Define the RMS value of an alternating quantity.

Answer: The RMS (effective) value is that steady DC value which produces the same heating effect in a resistance as the AC does. $V_{rms} = 0.707 V_m$.

Q11. Define form factor and peak factor.

Answer: Form factor = RMS value / average value = 1.11. Peak factor = maximum value / RMS value = 1.414.

Q12. What is a phasor?

Answer: A phasor is a line of definite length rotating anticlockwise at a constant angular velocity ω . Its length represents the maximum value and its angle the phase of the sinusoid.

Q13. Name the four forms of representing a phasor.

Answer: Rectangular ($a + jb$), trigonometrical ($V(\cos \theta + j \sin \theta)$), polar ($V\angle\theta$) and exponential ($Ve^{j\theta}$).

Q14. What is the j operator?

Answer: $j = \sqrt{-1}$ is called the operator. Multiplying a phasor by j rotates it through $+90^\circ$ anticlockwise, and by $-j$ through -90° .

Q15. In a pure resistance, what is the phase relationship between voltage and current?

Answer: The current is in phase with the voltage; the phase difference is 0° and the power factor is 1.

Q16. In a pure inductor, what is the phase relationship and reactance?

Answer: The current lags the voltage by 90° ; the inductive reactance is $X_L = 2\pi fL$ and the power factor is zero.

Q17. In a pure capacitor, what is the phase relationship and reactance?

Answer: The current leads the voltage by 90° ; the capacitive reactance is $X_C = 1/(2\pi fC)$ and the power factor is zero.

Q18. Write the impedance of an R-L series circuit.

Answer: $Z = R + jX_L$, so $|Z| = \sqrt{R^2 + X_L^2}$ and the current lags the voltage by $\phi = \tan^{-1}(X_L/R)$.

Q19. Write the impedance of an R-C series circuit.

Answer: $Z = R - jX_C$, so $|Z| = \sqrt{R^2 + X_C^2}$ and the current leads the voltage by $\phi = \tan^{-1}(X_C/R)$.

Q20. Write the impedance of an R-L-C series circuit.

Answer: $Z = R + j(X_L - X_C)$, so $|Z| = \sqrt{R^2 + (X_L - X_C)^2}$.

Q21. What is series resonance?

Answer: The condition in a series R-L-C circuit when $X_L = X_C$. The impedance is minimum ($Z = R$), the current is maximum, and the current and voltage are in phase.

Q22. Give the formula for the resonant frequency.

Answer: $f_0 = 1 / (2\pi\sqrt{LC})$.

Q23. What is parallel resonance?

Answer: The condition in a parallel circuit when the reactive currents cancel; the impedance is maximum and the line current is minimum.

Q24. Define real power, reactive power and apparent power.

Answer: Real power $P = VI \cos \phi$ (watts) is the power consumed; reactive power $Q = VI \sin \phi$ (VAR) is exchanged with L and C; apparent power $S = VI$ (VA) is their product.

Q25. What is the relation between P, Q and S?

Answer: $S = \sqrt{P^2 + Q^2}$, and the power factor $\cos \phi = P/S$.

Q26. Define power factor.

Answer: Power factor = $\cos \phi = P/S = R/|Z|$. It is lagging for inductive circuits and leading for capacitive circuits.

Q27. How is power measured in an AC circuit?

Answer: With a wattmeter (dynamometer type), which has a current coil in series and a potential coil in parallel with the load; it reads $V I \cos \phi$.

Q28. What is a polyphase system?

Answer: A system in which the AC sources operate at the same frequency but different phases. A three-phase system has three voltages displaced by 120° .

Q29. Write the equations of the three phase voltages.

Answer: $e_A = E_m \sin \omega t$, $e_B = E_m \sin(\omega t - 120^\circ)$, $e_C = E_m \sin(\omega t - 240^\circ)$; their sum is zero at every instant.

Q30. What is phase sequence?

Answer: The order in which the three phase voltages reach their maximum positive values. The usual sequence is RYB.

Q31. Write the phase and line relations for a star connection.

Answer: $V_L = \sqrt{3} V_{ph}$ and $I_L = I_{ph}$. With a neutral it is a 3-phase, 4-wire system.

Q32. Write the phase and line relations for a delta connection.

Answer: $V_L = V_{ph}$ and $I_L = \sqrt{3} I_{ph}$. No neutral exists, so only a 3-wire system is possible.

Q33. What is a balanced load?

Answer: A load in which the phase impedances are equal in magnitude and in phase. For a balanced load the three line currents sum to zero.

Q34. Give the formula for total power in a three-phase system.

Answer: $P = \sqrt{3} V_L I_L \cos \phi$; $Q = \sqrt{3} V_L I_L \sin \phi$; $S = \sqrt{3} V_L I_L$.

SECTION B — 5-Mark Questions

Q1. Explain an alternating quantity and its important terms (cycle, time period, frequency, amplitude, angular velocity).

Answer:

- An alternating voltage periodically changes its polarity; an alternating current changes its direction.
- Cycle: one complete set of positive and negative values. Time period T : time for one cycle.
- Frequency $f = 1/T$ cycles per second (Hz). Amplitude V_m : maximum value.
- Angular velocity $\omega = 2\pi/T = 2\pi f$ rad/s.

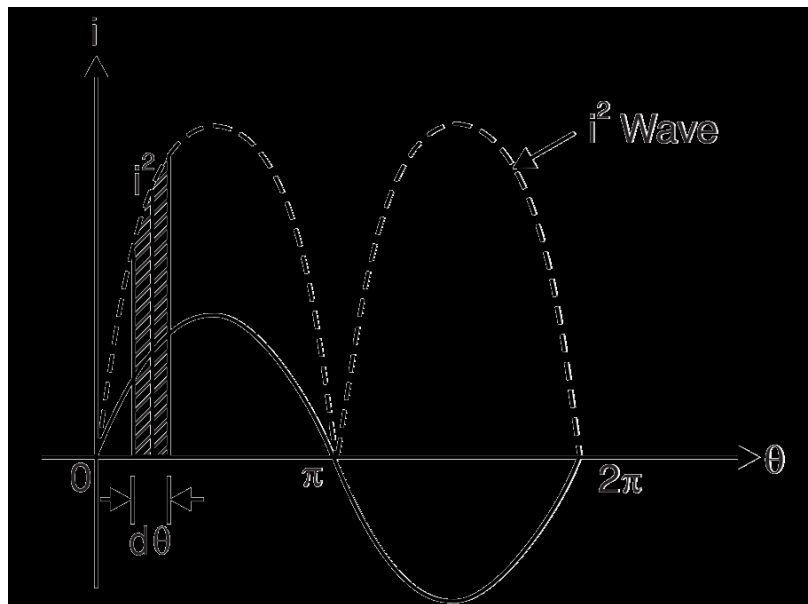


Figure 1: The sine waveform: cycle, peak and time period.

Q2. Explain the peak, average and RMS values of an alternating quantity.

Answer:

The peak value is the maximum value V_m . The average value over a full cycle is zero, but the half-cycle average is $0.637 V_m$. The RMS (effective) value is the steady DC value producing the same heating effect, equal to $0.707 V_m$.

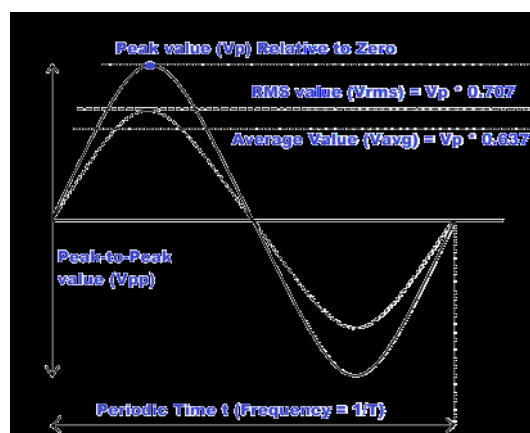


Figure 2: Peak, average and RMS values on the waveform.

Q3. Explain form factor and peak factor, and state their values for a sine wave.**Answer:**

Form factor = RMS value / average value = $0.707 V_m / 0.637 V_m = 1.11$.

Peak factor = maximum value / RMS value = $V_m / 0.707 V_m = 1.414$.

Q4. Explain the concept of a phasor and how a sinusoid is represented by a phasor.**Answer:**

A phasor is a line of definite length rotating anticlockwise at angular velocity ω ; its length represents the maximum value and its angle the phase. A sinusoid $v(t) = V_m \cos(\omega t + \phi)$ is represented by the phasor $V = V_m \angle \phi$, since $v(t) = \text{Re}(V e^{j\omega t})$.

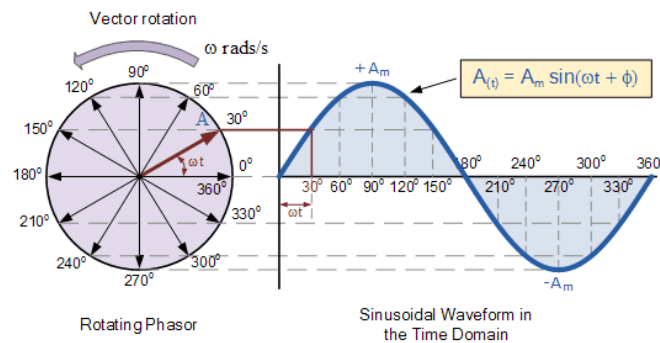


Figure 3: Rectangular components of a phasor.

Q5. Explain the four forms of representing a phasor.**Answer:**

- Rectangular: $V = a + jb$, with $V = \sqrt{a^2 + b^2}$ and $\theta = \tan^{-1}(b/a)$.
- Trigonometrical: $V = V(\cos \theta + j \sin \theta)$.
- Polar: $V = V \angle \theta$.
- Exponential: $V = V e^{j\theta}$ (Euler's equation).

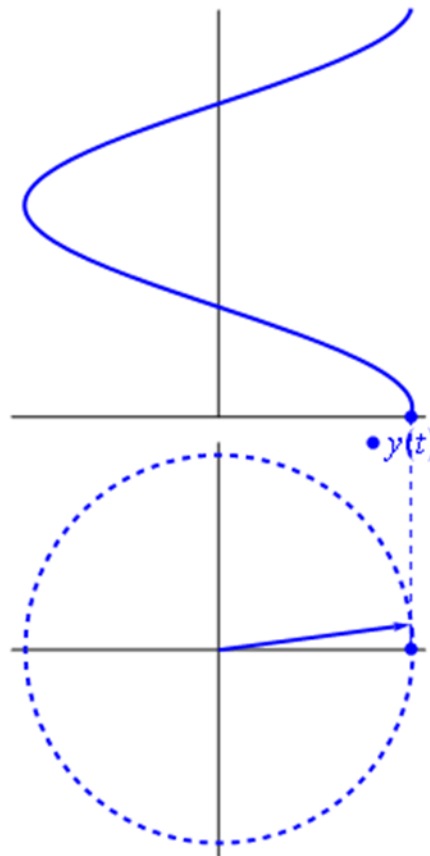


Figure 4: Polar and trigonometrical representation.

Q6. Explain the behaviour of a pure resistance in an AC circuit.

Answer:

With $v = V_m \sin \omega t$, the current $i = (V_m/R) \sin \omega t$ is in phase with the voltage. Impedance $Z = R$, power factor = 1, and the average power $P = VI = I^2R$.

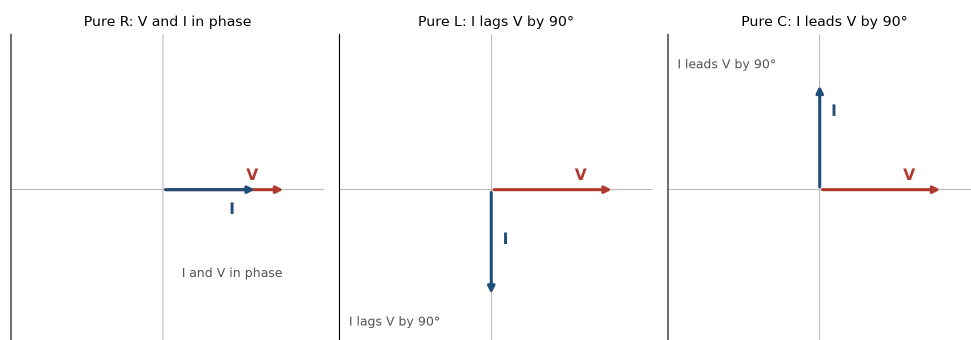


Figure 5: R, L and C phasor diagrams.

Q7. Explain the behaviour of a pure inductor and a pure capacitor in an AC circuit.

Answer:

Pure inductor: the current lags the voltage by 90° , $X_L = 2\pi fL$, power factor = 0, average power = 0. Pure capacitor: the current leads the voltage by 90° , $X_C = 1/(2\pi fC)$, power factor = 0, average power = 0.

Q8. Explain the R-L series circuit with its phasor diagram and impedance triangle.

Answer:

The same current flows through R and L. V_R is in phase with I and V_L leads by 90° , so the applied voltage V is the phasor sum. $Z = R + jX_L$, $|Z| = \sqrt{R^2 + X_L^2}$, and the current lags the voltage by $\phi = \tan^{-1}(X_L/R)$.

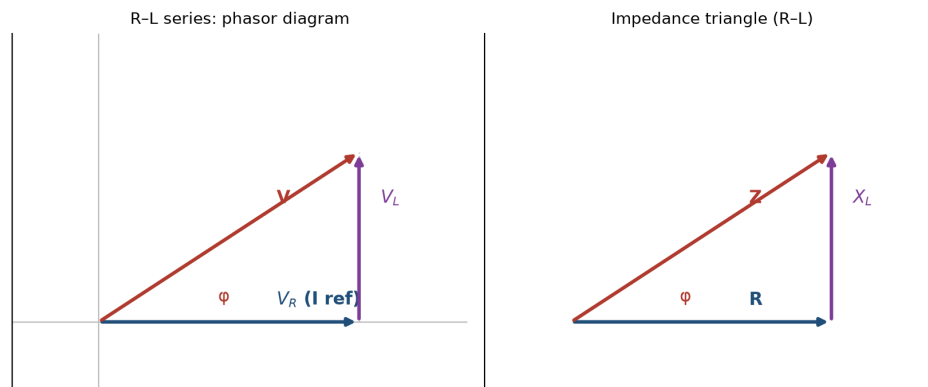


Figure 6: R-L phasor diagram and impedance triangle.

Q9. Explain the R-L-C series circuit.

Answer:

The applied voltage is the phasor sum of V_R , V_L and V_C . The net reactance is $(X_L - X_C)$, so $Z = R + j(X_L - X_C)$ and $|Z| = \sqrt{R^2 + (X_L - X_C)^2}$. The circuit is inductive, capacitive or resonant depending on whether $X_L > X_C$, $X_C > X_L$ or $X_L = X_C$.

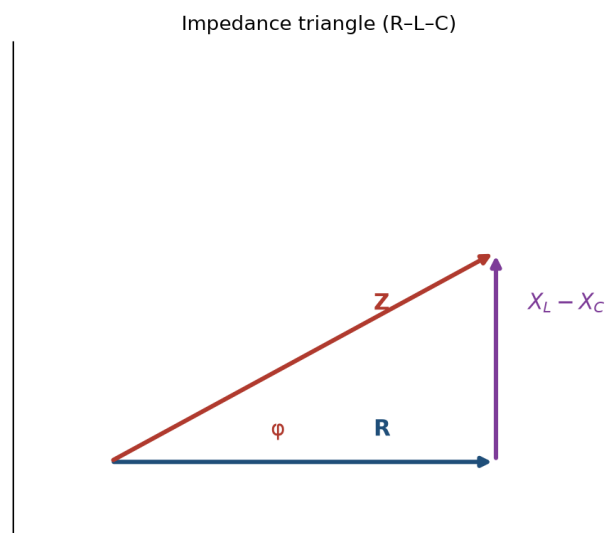


Figure 7: Impedance triangle of the R-L-C circuit.

Q10. Explain series resonance and its important properties.

Answer:

Series resonance occurs when $X_L = X_C$, giving $f_0 = 1/(2\pi\sqrt{LC})$. At resonance $Z = R$ is minimum, the current is maximum (V/R), the current and voltage are in phase (power factor unity), and the voltages across L and C can greatly exceed the supply voltage.

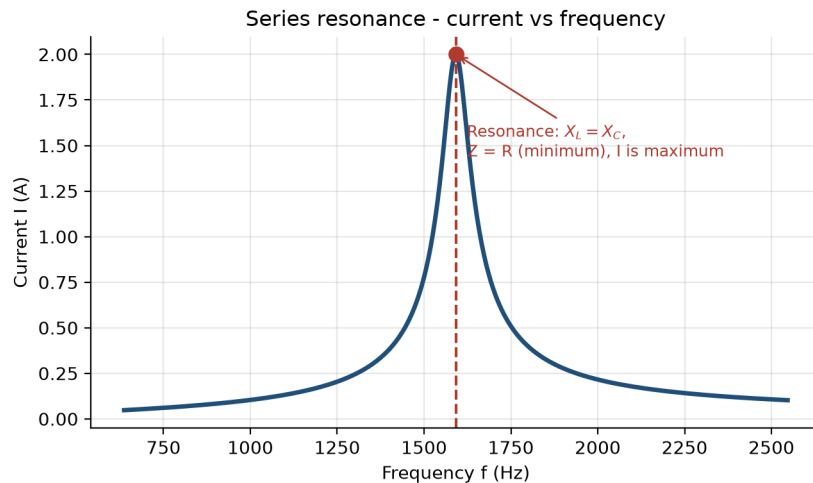


Figure 8: Current vs frequency — maximum at resonance.

Q11. Explain parallel resonance.

Answer:

In a parallel circuit, resonance occurs when the reactive currents cancel. The impedance is maximum (ideally infinite) and the line current is minimum; the supply current and voltage are in phase. For an ideal parallel circuit the resonant frequency is also $f_0 = 1/(2\pi\sqrt{LC})$.

Q12. Explain real power, reactive power, apparent power and the power triangle.

Answer:

Real power $P = VI \cos \phi$ (W) is actually consumed; reactive power $Q = VI \sin \phi$ (VAR) is exchanged with L and C; apparent power $S = VI$ (VA) is the product of RMS voltage and current. They form the power triangle with $S = \sqrt{P^2 + Q^2}$, and $\cos \phi = P/S$.

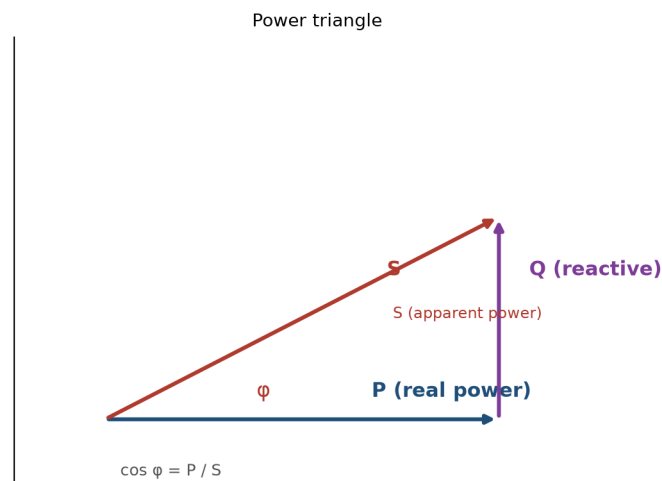


Figure 9: The power triangle.

Q13. Explain how power is measured in an AC circuit using a wattmeter.

Answer:

A wattmeter is a dynamometer-type instrument with two coils: the current coil in series with the load, and the potential coil in parallel with the load through a high resistance. Its deflection is proportional to $V I \cos \phi$, so it directly reads real power. Reading = deflection $\times$ multiplying factor.

Q14. Explain the generation of three-phase power and the advantages of a three-phase system.

Answer:

Three equal voltages displaced by 120° are generated: $e_A = E_m \sin \omega t$, $e_B = E_m \sin(\omega t - 120^\circ)$, $e_C = E_m \sin(\omega t - 240^\circ)$; their sum is zero. Advantages: constant power, greater output for the same size, cheaper motors and transmission, and a rotating magnetic field.

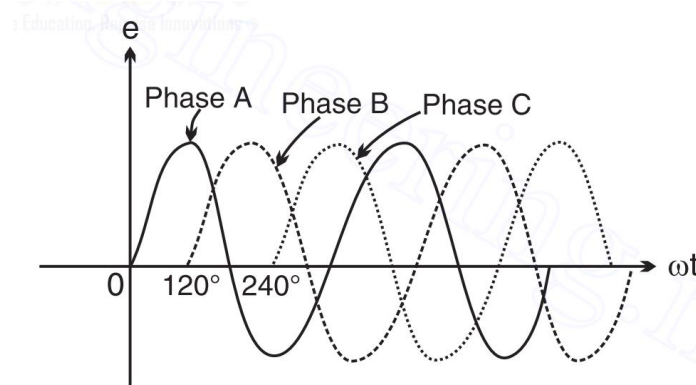


Figure 10: The three phase voltages displaced by 120° .

Q15. Explain the star connection with its phase and line relations.

Answer:

The three windings are joined at a common neutral point. With a neutral it is a 3-phase, 4-wire system; without it, a 3-wire system. $V_L = \sqrt{3} V_{ph}$ and $I_L = I_{ph}$.

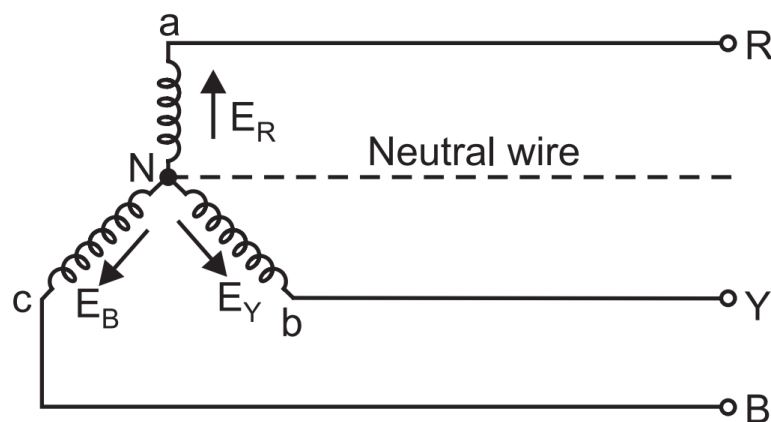


Figure 11: Star-connected system.

Q16. Explain the delta connection with its phase and line relations.

Answer:

The three windings are connected end-to-end in a closed loop. No neutral exists, so only a 3-wire system is possible. $V_L = V_{ph}$ and $I_L = \sqrt{3} I_{ph}$.

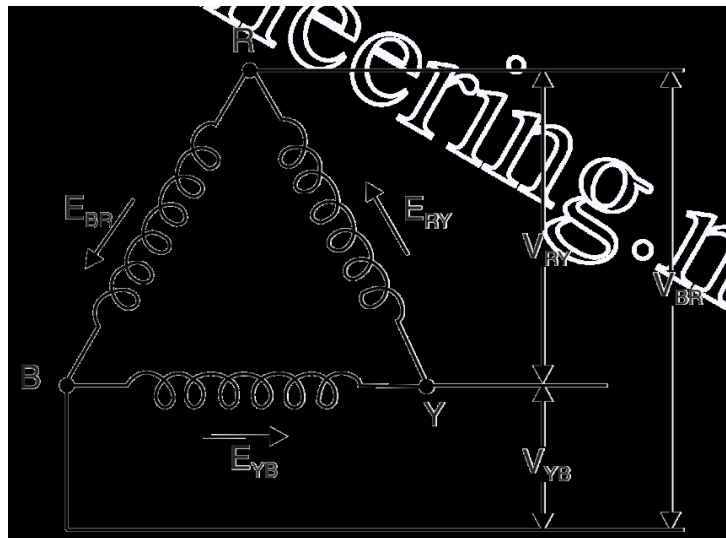


Figure 12: Delta-connected system.

SECTION C — 7-Mark Questions

Q1. Explain an alternating quantity, its important terms and the average, RMS, form and peak factors.

Answer:

An alternating quantity periodically reverses. Its terms are cycle, time period T , frequency $f = 1/T$, amplitude V_m , and angular velocity $\omega = 2\pi f$. The average (half-cycle) value is $0.637 V_m$, the RMS value is $0.707 V_m$, the form factor is 1.11 and the peak factor is 1.414.

The domestic supply 230 V, 50 Hz is an RMS value, so the peak is 325 V and $v = 230\sqrt{2} \sin(314t)$.

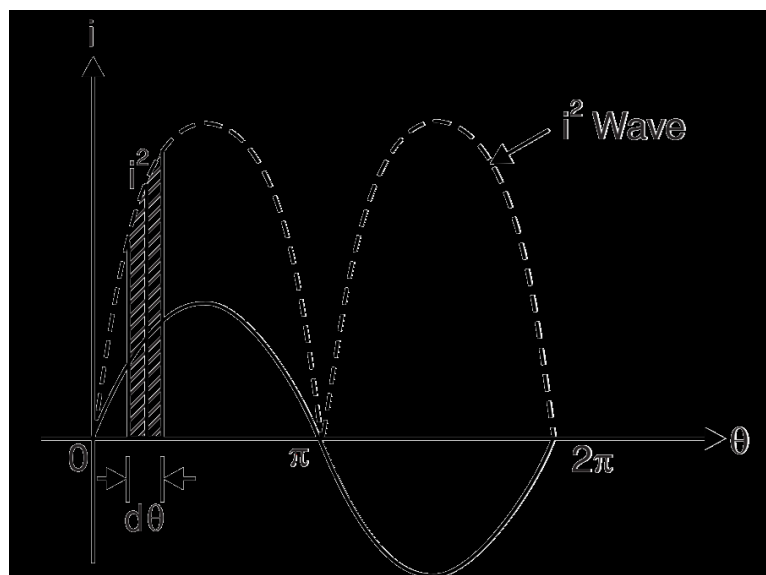


Figure 13: The sine waveform.

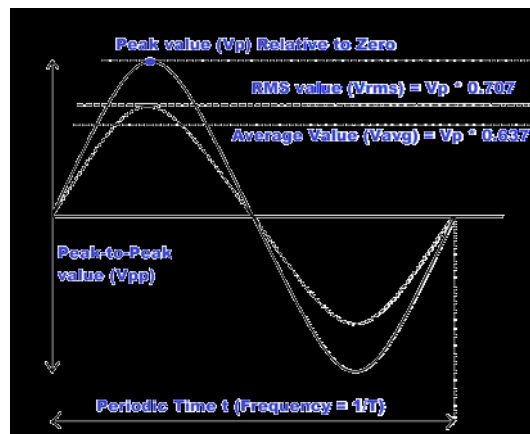


Figure 14: Peak, average and RMS values.

Q2. Explain phasors and the representation of sinusoids by phasors, with the four forms.

Answer:

A phasor is a rotating line whose length is the maximum value and whose angular velocity is ω . A sinusoid $v(t) = V_m \cos(\omega t + \phi)$ corresponds to the phasor $V = V_m \angle \phi$. The four forms are rectangular ($a + jb$), trigonometrical, polar and exponential. Basic operations (addition, subtraction, multiplication, division) are performed as for complex numbers.

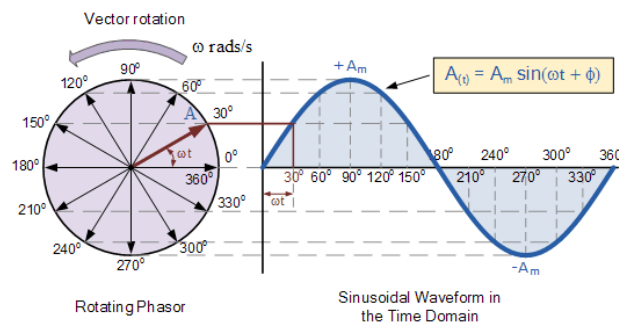


Figure 15: Rectangular components.

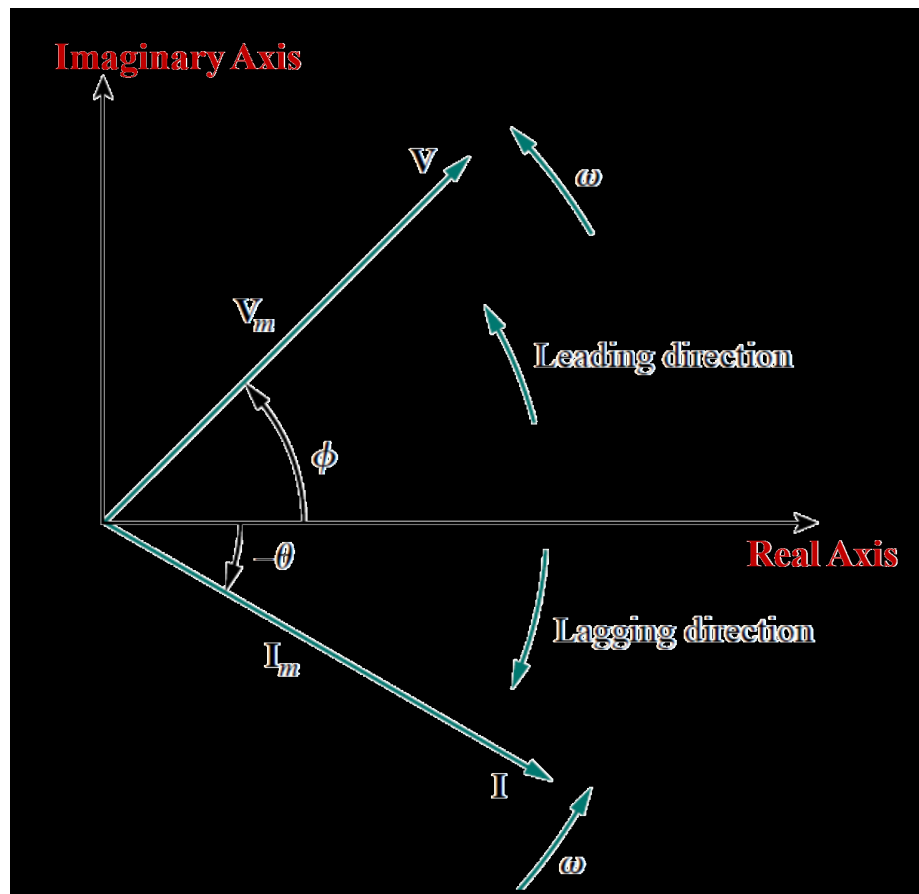


Figure 16: Phasor diagram and forms.

Q3. Explain the behaviour of R, L and C in an AC circuit with phasor diagrams.**Answer:****Pure R:** current in phase with voltage; $Z = R$; power factor 1; average power = VI .**Pure L:** current lags by 90° ; $X_L = 2\pi fL$; power factor 0; average power 0.**Pure C:** current leads by 90° ; $X_C = 1/(2\pi fC)$; power factor 0; average power 0.

The j operator explains this: multiplying by j rotates a phasor by $+90^\circ$, so $Z_L = j\omega L$ and $Z_C = -j/(\omega C)$.

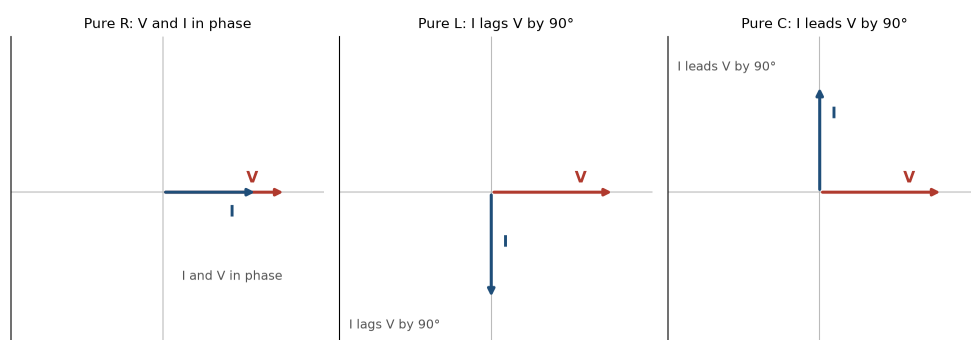


Figure 17: Phasor diagrams for R, L and C.

Q4. Explain the R-L, R-C and R-L-C series circuits in detail with impedance triangles.**Answer:****R-L:** $Z = R + jX_L$, $|Z| = \sqrt{R^2 + X_L^2}$; current lags by $\phi = \tan^{-1}(X_L/R)$.

R-C: $Z = R - jX_C$, $|Z| = \sqrt{R^2 + X_C^2}$; current leads by $\phi = \tan^{-1}(X_C/R)$.

R-L-C: $Z = R + j(X_L - X_C)$, $|Z| = \sqrt{R^2 + (X_L - X_C)^2}$; the circuit is inductive, capacitive or resonant accordingly.

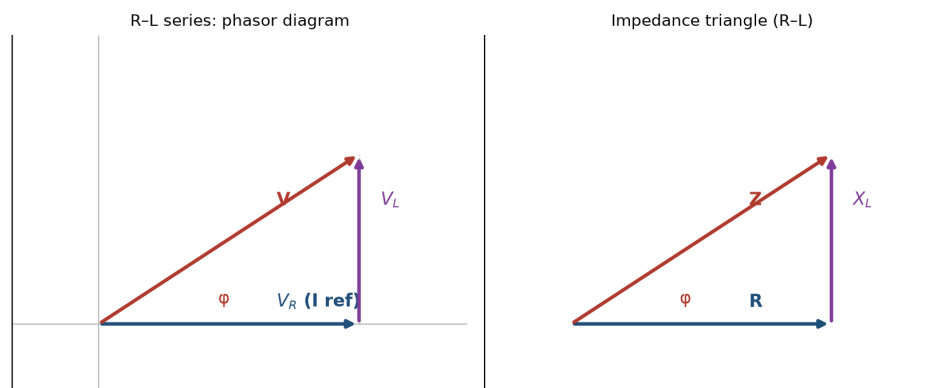


Figure 18: R-L phasor diagram and impedance triangle.

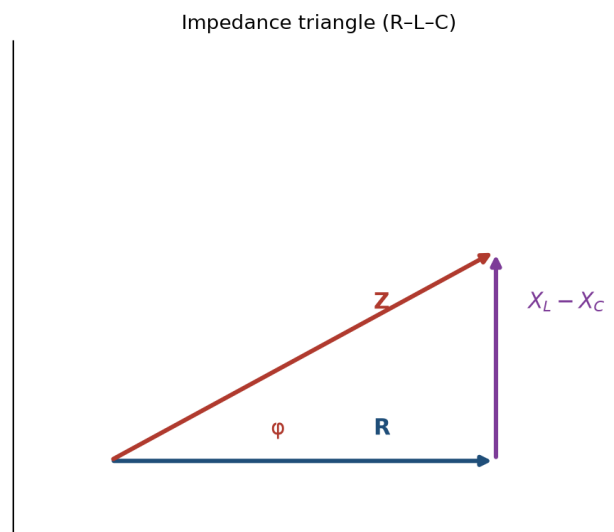


Figure 19: R-L-C impedance triangle.

Q5. Explain series and parallel resonance in detail with the resonant frequency and properties.

Answer:

Series resonance: $X_L = X_C$, $f_0 = 1/(2\pi\sqrt{LC})$; $Z = R$ is minimum, the current is maximum, the power factor is unity and the voltage across L and C is magnified. Quality factor $Q = (1/R)\sqrt{L/C}$ and bandwidth $= f_0/Q$.

Parallel resonance: the reactive currents cancel; the impedance is maximum and the line current minimum; the current and voltage are in phase.

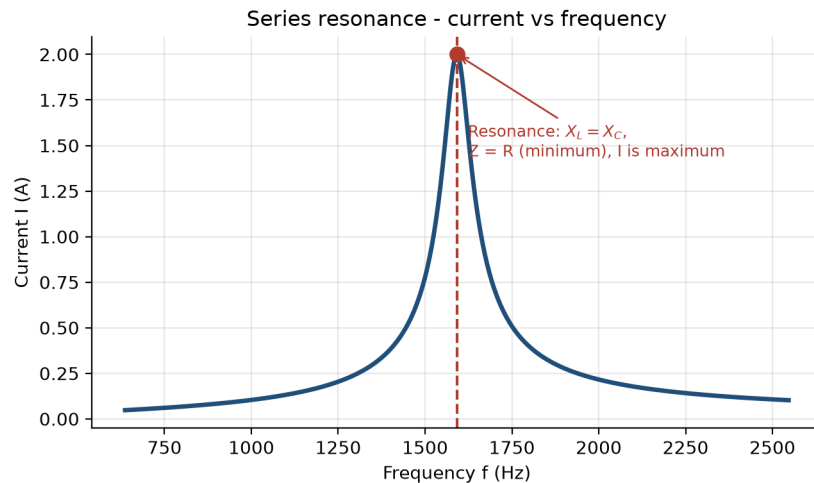


Figure 20: Series resonance curve.

Q6. Explain power in AC circuits, the power triangle and power factor in detail.**Answer:**

Real power $P = VI \cos \phi$ (watts), reactive power $Q = VI \sin \phi$ (VAR), apparent power $S = VI$ (VA). They form the power triangle with $S = \sqrt{P^2 + Q^2}$ and power factor $\cos \phi = P/S = R/|Z|$. For a pure resistance the power factor is 1; for a pure L or C it is zero. Power factor is lagging for inductive circuits and leading for capacitive circuits.

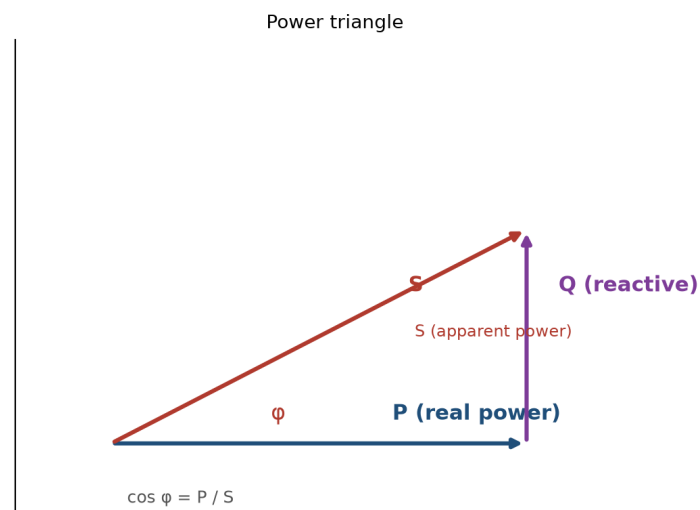


Figure 21: The power triangle.

Q7. Explain how power is measured in AC circuits, and explain the wattmeter.**Answer:**

Power is measured with a wattmeter of the dynamometer type. It has a current coil (CC) connected in series with the load and a potential coil (PC) connected in parallel through a high resistance. The deflection is proportional to $V I \cos \phi$, so the instrument reads real power directly. The reading is multiplied by the multiplying factor = (voltage range $\times$ current range)/(full-scale deflection).

Q8. Explain the generation of three-phase power, the phase sequence and the advantages of three-phase systems.

Answer:

Three equal voltages displaced by 120° are generated. The resultant is zero at every instant. The phases are named R, Y, B with the sequence RYB. Advantages: constant power, greater output for the same size, cheaper motors, economical transmission, and a rotating magnetic field.

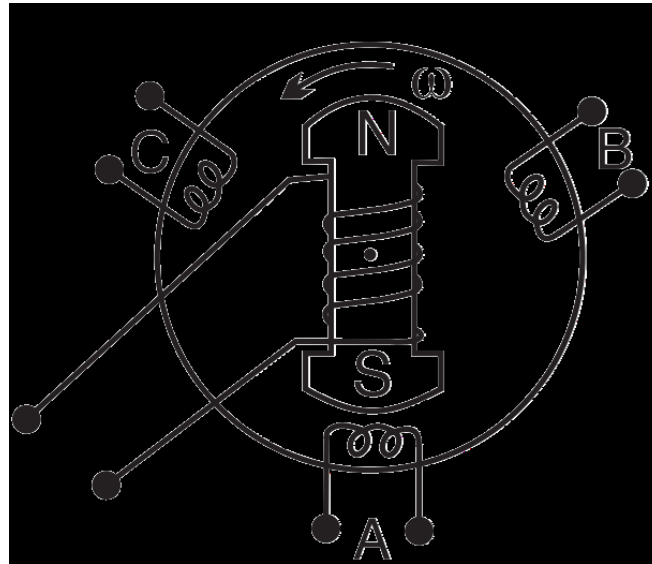


Figure 22: Three-phase generator.

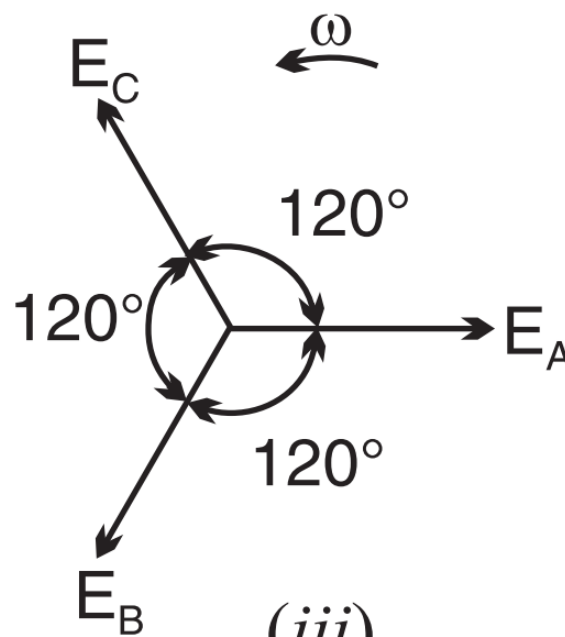


Figure 23: Three-phase phasor diagram.

Q9. Explain the star and delta connections of a three-phase system with the phase-line relations and a phasor diagram.

Answer:

Star: windings joined at a neutral; $V_L = \sqrt{3} V_{ph}$, $I_L = I_{ph}$; 4-wire (with neutral) or 3-wire (without neutral).

Delta: windings connected end-to-end; $V_L = V_{ph}$, $I_L = \sqrt{3} I_{ph}$; no neutral, only 3-wire. For a balanced load the phase impedances are equal, with $Z_\Delta = 3 Z_Y$, and total power $P = \sqrt{3} V_L I_L \cos \phi$.

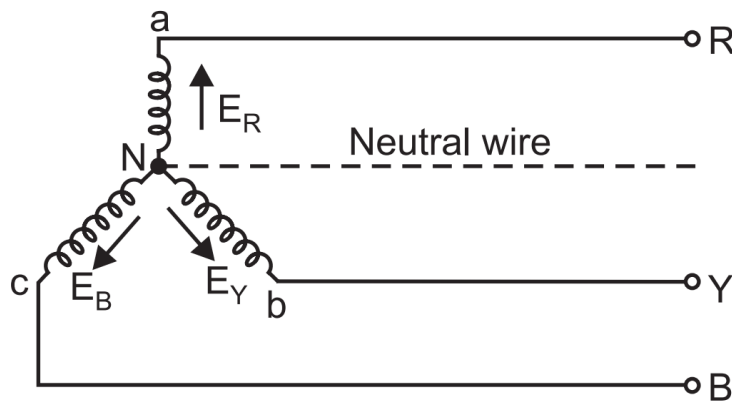


Figure 24: Star connection.

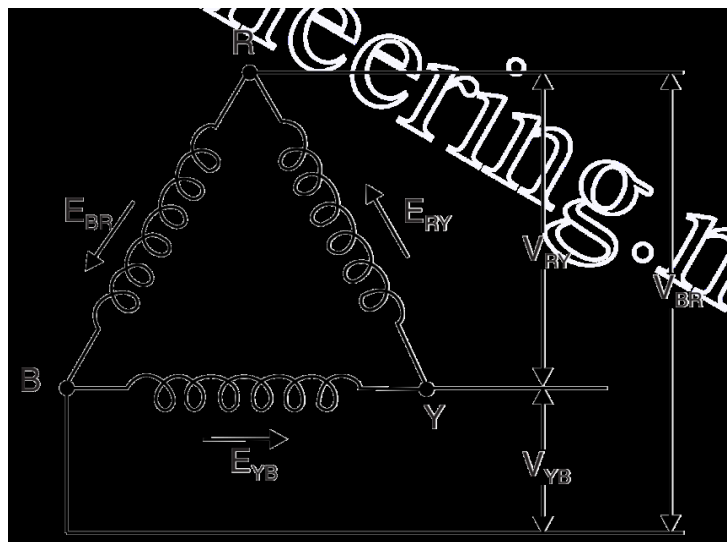


Figure 25: Delta connection.

Q10. Explain the three-phase power relations and the advantages of a three-phase system.

Answer:

Total three-phase power = 3 × power per phase. In terms of line quantities: $P = \sqrt{3} V_L I_L \cos \phi$, $Q = \sqrt{3} V_L I_L \sin \phi$ and $S = \sqrt{3} V_L I_L = \sqrt{(P^2 + Q^2)}$. For a balanced system the three line currents sum to zero.

- Constant power at every instant.
- Greater output for the same machine size and weight.
- Cheaper motors and economical power transmission.
- Sets up a rotating magnetic field in stationary windings.

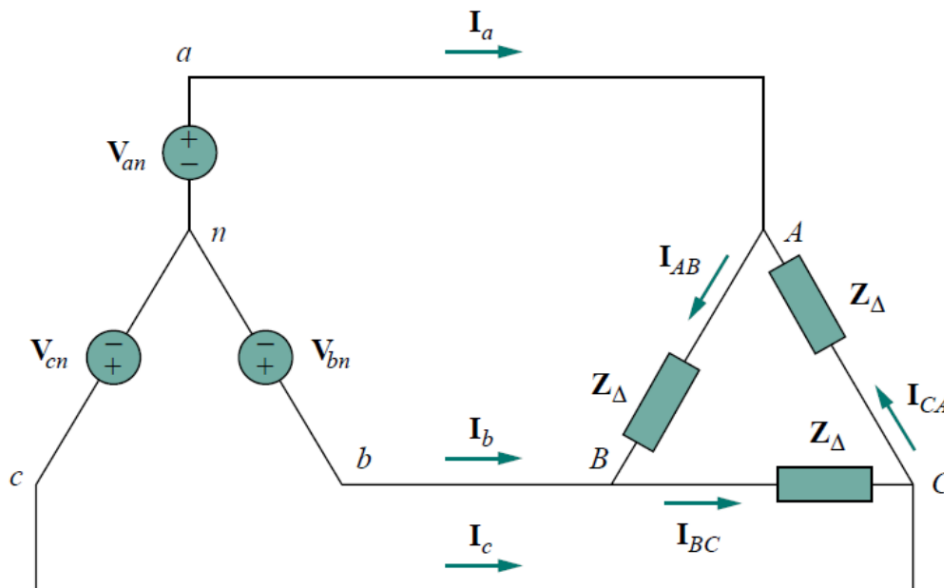


Figure 26: Source-load connection combinations.

SECTION D — Numerical Problems (Solved)

Every numerical below is solved step by step, with the circuit or phasor diagram where needed.

N1. AC Values from the Equation

Problem: A sinusoidal voltage is $v(t) = 100 \sin(314t)$. Find the peak value, RMS value, average value, frequency and time period.

Solution:

Peak value $V_m = 100 \text{ V}$.

RMS value $V_{\text{rms}} = 0.707 \times 100 = 70.7 \text{ V}$; average value $V_{\text{av}} = 0.637 \times 100 = 63.7 \text{ V}$.

Angular velocity $\omega = 314 \text{ rad/s}$, so frequency $f = \omega/2\pi = 314/(2\pi) = 50 \text{ Hz}$.

Time period $T = 1/f = 1/50 = 0.02 \text{ s}$.

Answer: $V_m = 100 \text{ V}$, $V_{\text{rms}} = 70.7 \text{ V}$, $V_{\text{av}} = 63.7 \text{ V}$, $f = 50 \text{ Hz}$, $T = 0.02 \text{ s}$.

N2. Form Factor and Peak Factor

Problem: A sinusoidal voltage has a peak value of 200 V . Find its RMS value, average value, form factor and peak factor.

Solution:

$V_{\text{rms}} = 0.707 \times 200 = 141.4 \text{ V}$.

$V_{\text{av}} = 0.637 \times 200 = 127.4 \text{ V}$.

Form factor $= V_{\text{rms}}/V_{\text{av}} = 141.4/127.4 = 1.11$.

Peak factor $= V_m/V_{\text{rms}} = 200/141.4 = 1.414$.

Answer: $V_{\text{rms}} = 141.4 \text{ V}$, $V_{\text{av}} = 127.4 \text{ V}$, form factor $= 1.11$, peak factor $= 1.414$.

N3. Phasor Representation and Addition

Problem: Express $v = 50 \sin(\omega t + 30^\circ)$ as a phasor in rectangular form, and add the phasors $V_1 = 30 \angle 0^\circ$ and $V_2 = 40 \angle 90^\circ$.

Solution:

The phasor is $V = 50\angle 30^\circ = 50(\cos 30^\circ + j \sin 30^\circ) = 43.3 + j25$.

For the addition: $V_1 = 30\angle 0^\circ = 30 + j0$ and $V_2 = 40\angle 90^\circ = 0 + j40$.

$V_1 + V_2 = 30 + j40$; magnitude $= \sqrt{30^2 + 40^2} = 50$; angle $= \tan^{-1}(40/30) = 53.13^\circ$.

Answer: $V = 43.3 + j25$; $V_1 + V_2 = 50\angle 53.13^\circ$.

N4. Pure Resistance

Problem: A resistance of $46\ \Omega$ is connected to a 230 V, 50 Hz supply. Find the current and the power consumed.

Solution:

For a pure resistance the current is in phase with the voltage.

$$I = V/R = 230/46 = 5\text{ A.}$$

$$\text{Power } P = VI = 230 \times 5 = 1150\text{ W (also } I^2R = 5^2 \times 46 = 1150\text{ W).}$$

Answer: $I = 5\text{ A}$, $P = 1150\text{ W}$.

N5. Pure Inductance

Problem: An inductor of 0.1 H is connected to a 230 V, 50 Hz supply. Find the inductive reactance, current and reactive power.

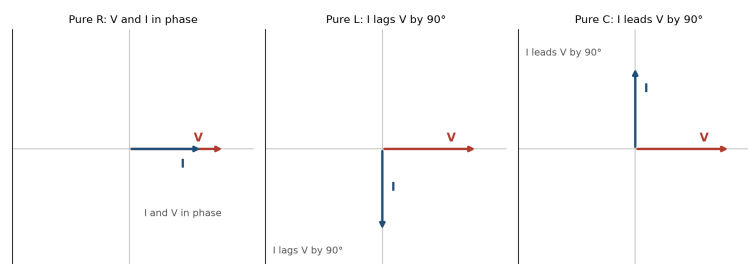


Figure 27: Figure for N5.

Solution:

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.1 = 31.4\ \Omega.$$

$$I = V/X_L = 230/31.4 = 7.32\text{ A (current lags the voltage by } 90^\circ).$$

$$\text{Reactive power } Q = VI = 230 \times 7.32 = 1684\text{ VAR; real power } P = 0.$$

Answer: $X_L = 31.4\ \Omega$, $I = 7.32\text{ A}$, $Q = 1684\text{ VAR}$, $P = 0$.

N6. Pure Capacitance

Problem: A capacitor of $50\ \mu\text{F}$ is connected to a 230 V, 50 Hz supply. Find the capacitive reactance, current and reactive power.

Solution:

$$X_C = 1/(2\pi fC) = 1/(2\pi \times 50 \times 50 \times 10^{-6}) = 63.66\ \Omega.$$

$$I = V/X_C = 230/63.66 = 3.61\text{ A (current leads the voltage by } 90^\circ).$$

$$\text{Reactive power } Q = VI = 230 \times 3.61 = 831\text{ VAR; real power } P = 0.$$

Answer: $X_C = 63.66\ \Omega$, $I = 3.61\text{ A}$, $Q = 831\text{ VAR}$.

N7. R-L Series Circuit

Problem: A coil of $R = 6\ \Omega$ and $L = 0.05\text{ H}$ is connected to a 230 V, 50 Hz supply. Find the impedance, current, power factor and power.

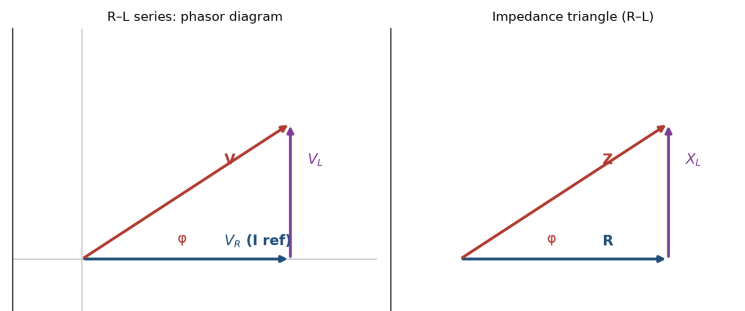


Figure 28: Figure for N7.

Solution:

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.05 = 15.71 \, \Omega.$$

$$|Z| = \sqrt{R^2 + X_L^2} = \sqrt{6^2 + 15.71^2} = 16.82 \, \Omega.$$

$$I = V/|Z| = 230/16.82 = 13.68 \, \text{A}.$$

$$\phi = \tan^{-1}(15.71/6) = 69.1^\circ \text{ (current lags); power factor} = \cos 69.1^\circ = 0.357 \text{ lagging}.$$

$$\text{Power } P = VI \cos \phi = 230 \times 13.68 \times 0.357 = 1123 \, \text{W (also } I^2 R = 13.68^2 \times 6 = 1123 \, \text{W)}.$$

Answer: $|Z| = 16.82 \, \Omega$, $I = 13.68 \, \text{A}$, $\text{pf} = 0.357$ lagging, $P = 1123 \, \text{W}$.

N8. R-C Series Circuit

Problem: A resistance of $10 \, \Omega$ is in series with a capacitor of $100 \, \mu\text{F}$ across a $230 \, \text{V}$, $50 \, \text{Hz}$ supply. Find the impedance, current and power factor.

Solution:

$$X_C = 1/(2\pi fC) = 1/(2\pi \times 50 \times 100 \times 10^{-6}) = 31.83 \, \Omega.$$

$$|Z| = \sqrt{R^2 + X_C^2} = \sqrt{10^2 + 31.83^2} = 33.36 \, \Omega.$$

$$I = 230/33.36 = 6.89 \, \text{A}.$$

$$\phi = \tan^{-1}(31.83/10) = 72.6^\circ \text{ (current leads); power factor} = \cos 72.6^\circ = 0.30 \text{ leading}.$$

Answer: $|Z| = 33.36 \, \Omega$, $I = 6.89 \, \text{A}$, $\text{pf} = 0.30$ leading.

N9. R-L-C Series Circuit

Problem: An R-L-C series circuit has $R = 10 \, \Omega$, $L = 0.1 \, \text{H}$ and $C = 50 \, \mu\text{F}$, connected to a $230 \, \text{V}$, $50 \, \text{Hz}$ supply. Find the impedance, current and power factor.

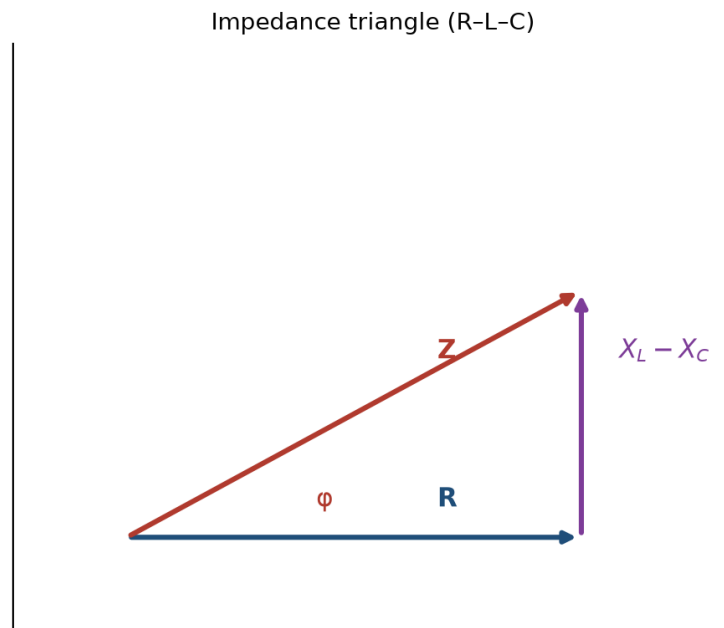


Figure 29: Figure for N9.

Solution:

$$X_L = 2\pi fL = 31.42 \, \Omega; X_C = 1/(2\pi fC) = 63.66 \, \Omega.$$

$$\text{Net reactance} = X_L - X_C = 31.42 - 63.66 = -32.24 \, \Omega \text{ (capacitive).}$$

$$|Z| = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{10^2 + 32.24^2} = 33.75 \, \Omega.$$

$$I = 230/33.75 = 6.82 \, \text{A}.$$

$$\phi = \tan^{-1}(-32.24/10) = -72.8^\circ \text{ (current leads); power factor} = \cos 72.8^\circ = 0.296 \text{ leading.}$$

Answer: $|Z| = 33.75 \, \Omega$, $I = 6.82 \, \text{A}$, $\text{pf} = 0.296 \text{ leading}$.

N10. Series Resonance

Problem: An R-L-C series circuit has $R = 10 \, \Omega$, $L = 10 \, \text{mH}$ and $C = 1 \, \mu\text{F}$. Find the resonant frequency, the quality factor and the current at resonance for a 10 V supply.

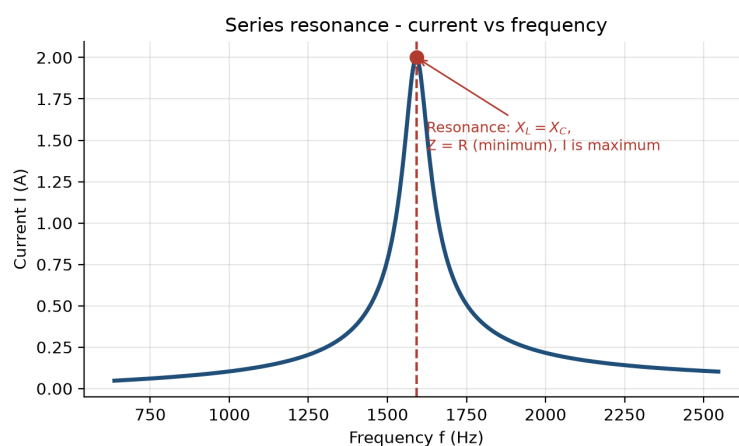


Figure 30: Figure for N10.

Solution:

$$f_0 = 1/(2\pi\sqrt{LC}) = 1/(2\pi\sqrt{0.01 \times 10^{-6}}) = 1591.5 \, \text{Hz}.$$

$$\text{Quality factor } Q = (1/R)\sqrt{L/C} = (1/10)\sqrt{0.01/10^{-6}} = (1/10)\sqrt{10000} = 10.$$

At resonance $X_L = X_C$, so $Z = R = 10 \Omega$ and $I = V/R = 10/10 = 1 \text{ A}$ (maximum).

Answer: $f_0 = 1591.5 \text{ Hz}$, $Q = 10$, $I = 1 \text{ A}$.

N11. Power and Power Factor

Problem: A circuit draws 10 A at 230 V with a power factor of 0.8 lagging. Find the real power, reactive power and apparent power.

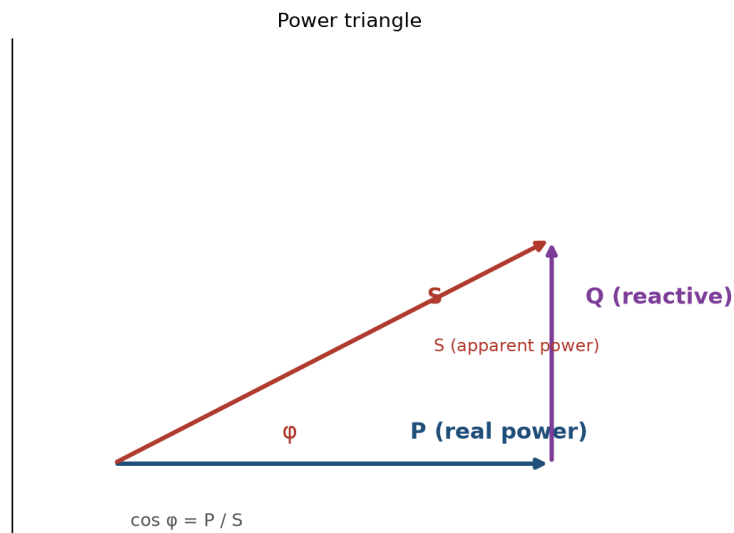


Figure 31: Figure for N11.

Solution:

Real power $P = VI \cos \phi = 230 \times 10 \times 0.8 = 1840 \text{ W}$.

$\phi = \cos^{-1}(0.8) = 36.87^\circ$, so $\sin \phi = 0.6$; reactive power $Q = VI \sin \phi = 230 \times 10 \times 0.6 = 1380 \text{ VAR}$.

Apparent power $S = VI = 230 \times 10 = 2300 \text{ VA}$ (check: $\sqrt{(1840^2 + 1380^2)} = 2300 \text{ VA}$).

Answer: $P = 1840 \text{ W}$, $Q = 1380 \text{ VAR}$, $S = 2300 \text{ VA}$.

N12. Three-Phase Balanced Load

Problem: A Y-connected balanced three-phase generator with a phase voltage of 220 V supplies a balanced Y-connected load of $(10 + j5) \Omega$ per phase. Find the line voltage, phase current, line current and total power.

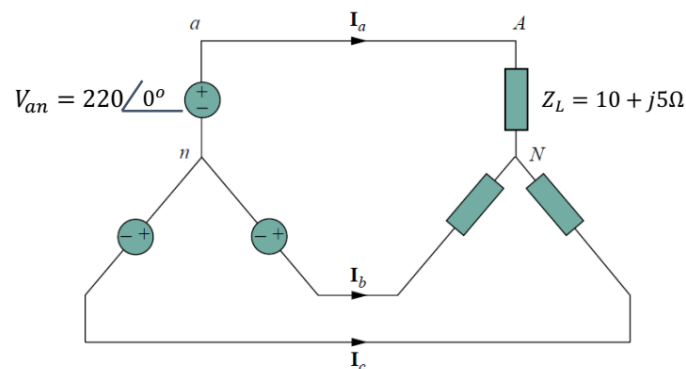


Figure 32: Figure for N12.

Solution:

Line voltage $V_L = \sqrt{3} \times V_{ph} = \sqrt{3} \times 220 = 381 \text{ V}$.

Impedance per phase $|Z| = \sqrt{10^2 + 5^2} = 11.18 \Omega$.

Phase current $I_{ph} = V_{ph}/|Z| = 220/11.18 = 19.68 \text{ A}$; for a star connection $I_L = I_{ph} = 19.68 \text{ A}$.

$\phi = \tan^{-1}(5/10) = 26.57^\circ$.

Real power $P = 3 I^2 R = 3 \times 19.68^2 \times 10 = 11619 \text{ W}$; reactive power $Q = 3 I^2 X = 3 \times 19.68^2 \times 5 = 5809 \text{ VAR}$.

Apparent power $S = \sqrt{P^2 + Q^2} = 12990 \text{ VA}$ (check: $\sqrt{3} V_L I_L = \sqrt{3} \times 381 \times 19.68 = 12990 \text{ VA}$).

Answer: $V_L = 381 \text{ V}$, $I_{ph} = I_L = 19.68 \text{ A}$, $P = 11619 \text{ W}$, $Q = 5809 \text{ VAR}$, $S = 12990 \text{ VA}$.